

Q.21 Find the acute angle between the lines whose slopes are $\frac{1}{\sqrt{3}}$ and $-\frac{1}{\sqrt{3}}$. (CO3)

Q.22 Find the radius and centre of given equation of circle. (CO3)

$$x^2 + y^2 + 2x + 4y - 4 = 0$$

Section-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Use Simpson rule to evaluate $\int_1^9 (x+2) dx$ by taking 8 equal sub-intervals. (CO4)

Q.24 Find the equation of the circle which passes through the following three points: (CO3)
(0,1), (1,0) & (-1,0).

Q.25 Prove that $\sin 10^\circ \sin 50^\circ \sin 60^\circ \sin 70^\circ = \frac{\sqrt{3}}{16}$. (CO1)

No. of Printed Pages : 4
Roll No.

222012

1st Year / Advance Diploma in Tool and Die Making Subject : Applied Mathematics

Time : 3 Hrs.

M.M. : 60

Section-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Value of i^{23} is (CO1)

- (a) i (b) -1
(c) $-i$ (d) 1

Q.2 In MATLAB, the command used to represent a "Not-a-Number" result is: (CO5)

- (a) null (b) empty
(c) nan (d) inf

Q.3 The angle _____ lies in the 4th quadrant. (CO2)

- (a) 195° (b) 150°
(c) 185° (d) 315°

Q.4 Centre of the circle $(x-4)^2 + (y-1)^2 = 1$ is (CO3)

- (a) (3,1) (b) (4,1)
(c) (-4,-1) (d) (-4,1)

Q.5 $\int (\sec x \tan x) dx = \underline{\hspace{2cm}}$ (CO4)

- (a) $\tan x + C$ (b) $\sec x + C$
 (c) $\cot x + C$ (d) none of these

Q.6 Which of the following is not an equation of circle? (CO3)

- (a) $\frac{x}{3} + \frac{y}{7} = 1$ (b) $(x-4)^2 + (y-1)^2 = 1$
 (c) $x^2 + 3y^2 = 27$ (d) $x^2 + (y-3)^2 = 7$

Section-B

Note: Objective type questions. All questions are compulsory. (6x1=6)

Q.7 Give an example of a scalar matrix. (CO1)

Q.8 What is the conjugate of the complex number $z = 2 + 4i$? (CO2)

Q.9 $\int e^{3x} dx = \underline{\hspace{2cm}}$ (CO4)

Q.10 Derivative of $\tan x$ with respect to x is _____. (CO4)

Q.11 Calculate the Median of the following data: 2, 3, 5, 4, 6, 1, 7. (CO3)

Q.12 Value of the determinant $\begin{vmatrix} 5 & -3 \\ 1 & 12 \end{vmatrix}$ is _____. (CO5)

(2)

222012

Section-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Solve $3x + y = 11$, $2x + 3y = 12$ using Cramer's rule. (CO1)

Q.14 If $z_1 = 2 + 3i$ & $z_2 = 1 + 2i$, then find the conjugate of $\frac{z_1}{z_2}$. (CO1)

Q.15 If $y = x^2 + \sin x + e^x$, then find $\frac{d^2y}{dx^2}$ (CO4)

Q.16 If $y = \frac{(2x-3)}{(x-1)}$, then find $\frac{dy}{dx}$. (CO4)

Q.17 Find the order and degree of the following differential equation: (CO4)

$$\tan x \left(\frac{d^5 y}{dx^5} \right)^{17} + \sin x \left(\frac{d^6 y}{dx^6} \right) + y \frac{dy}{dx} = y^7.$$

Q.18 Evaluate $\int_0^{\frac{\pi}{2}} \cos^{10} x \, dx$. (CO2)

Q.19 Calculate the Median of the following data: (CO1)

x	1	4	7	10	13	16
f	2	6	5	2	7	6

Q.20 Evaluate $\int x \sin x \, dx$. (CO4)

(3)

222012